

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET F — RAPID FIRE SPEED TEST (30 MINS)

CHAPTER 06: HOW FORCES AFFECT MOTION & LAWS OF MOTION
TIME ALLOWED: 30 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-F-06

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET F - RAPID FIRE SPEED TEST):

1. The question paper comprises **25 Rapid Fire Multiple Choice Questions** of **1 mark each**.
2. Designed for **Rapid Recall & High-Speed Accuracy**: Recommended target time is **30 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **SPEED CHALLENGE INSTRUCTIONS:** Answer promptly without looking at notes. Test your instant recall of formulas, scientific definitions, and core NCERT concepts!

Q.1 Which physical property of a body is a direct quantitative measure of its inertia?

- (A) Weight (B) Volume
(C) Mass (D) Velocity

Q.2 What is the SI unit of linear momentum?

- (A) $\text{N} \cdot \text{m}$ (B) $\text{kg} \cdot \text{m/s}$
(C) $\text{kg} \cdot \text{m/s}^2$ (D) Joule

Q.3 Which law of motion gives the quantitative mathematical formula for force ($F = ma$)?

- (A) Newton's First Law (B) Newton's Second Law
(C) Newton's Third Law (D) Law of Gravitation

Q.4 What net unbalanced force is required to accelerate a 5 kg mass at 4 m/s^2 ?

- (A) 1.25 N (B) 20 N
(C) 9 N (D) 24.5 N

Q.5 Newton's third law of motion states that action and reaction forces:

- (A) Act on the same body in the same direction (B) Act on two different bodies simultaneously in opposite directions
(C) Cancel each other to produce equilibrium (D) Occur with a noticeable time delay

Q.6 Why does a cricket fielder pull his hands backwards while catching a fast cricket ball?

- (A) To catch the ball higher (B) To increase the time duration over which momentum drops to zero, reducing the impact force on his hands
(C) To apply more force on the ball (D) To spin the ball

Q.7 When a passenger bus takes a sharp left turn, standing passengers tilt towards the right due to:

- | | |
|--------------------------|-----------------------|
| (A) Inertia of rest | (B) Inertia of motion |
| (C) Inertia of direction | (D) Centripetal pull |

Q.8 When a carpet is vigorously beaten with a stick, dust particles fall out due to:

- | | |
|-----------------------------|-------------------------|
| (A) Inertia of rest | (B) Inertia of motion |
| (C) Gravitational repulsion | (D) Magnetic attraction |

Q.9 A gun of mass 4 kg fires a bullet of mass 0.02 kg with velocity 400 m/s. What is the recoil velocity of the gun?

- | | |
|------------------------|-------------------------|
| (A) -2.0 m/s | (B) -4.0 m/s |
| (C) -0.5 m/s | (D) -20.0 m/s |

Q.10 What is the momentum of an object of mass m moving with velocity v ?

- | | |
|-----------------------|-------------|
| (A) $(mv)^2$ | (B) $m v^2$ |
| (C) $\frac{1}{2}mv^2$ | (D) mv |

Q.11 1 Newton is defined as the force that produces an acceleration of 1 m/s^2 on a mass of:

- | | |
|-----------|---------------|
| (A) 1 g | (B) 1 kg |
| (C) 10 kg | (D) 1 quintal |

Q.12 Rocket propulsion operates on the fundamental physical principle of:

- | | |
|---------------------------------|--|
| (A) Conservation of energy only | (B) Conservation of linear momentum and Newton's Third Law |
| (C) Bernoulli's principle | (D) Archimedes' principle |

Q.13 If the net force acting on a moving body is zero ($F_{\text{net}} = 0$), the body will:

- | | |
|-----------------------------|---|
| (A) Stop moving immediately | (B) Continue moving with constant velocity in a straight line |
| (C) Accelerate uniformly | (D) Move in a circle |

Q.14 Which force provides the necessary forward reaction push that enables a person to walk on ground?

- | | |
|-------------------------|--------------------------------------|
| (A) Gravitational force | (B) Friction force exerted by ground |
| (C) Centrifugal force | (D) Buoyant force |

Q.15 What is the momentum of a 1000 kg car parked at rest by the roadside?

- | | |
|-----------------|----------------|
| (A) 1000 kg·m/s | (B) Zero |
| (C) 9800 N | (D) 100 kg·m/s |

Q.16 Why are athletes in high-jump events provided with sand pits or foam mattresses to land on?

- | | |
|----------------------------------|---|
| (A) To increase landing velocity | (B) To increase the impact time, thereby reducing the rate of change of momentum and impact force |
| (C) To prevent bouncing | (D) To look modern |

Q.17 What is the product of force and the time interval over which it acts called?

- | | |
|-------------|--------------------|
| (A) Impulse | (B) Power |
| (C) Work | (D) Kinetic energy |

Q.18 In an isolated system with zero external force, what happens to total linear momentum during a collision?

- | | |
|-----------------------------------|---------------------|
| (A) It increases | (B) It decreases |
| (C) It remains strictly conserved | (D) It becomes zero |

Q.19 Which vehicle has greater inertia: a loaded truck of 10,000 kg or a scooter of 100 kg?

- (A) The scooter (B) The truck, because it has 100 times greater mass
(C) Both have identical inertia (D) Neither has inertia

Q.20 A force of 10 N acts on a body of mass 2 kg for 3 seconds. What is the change in momentum?

- (A) 15 kg·m/s (B) 30 kg·m/s
(C) 6 kg·m/s (D) 5 kg·m/s

Q.21 What force is needed to keep an object moving with constant velocity on a frictionless surface in deep space?

- (A) 10 N (B) Zero force
(C) Equal to its weight (D) Infinitely large force

Q.22 When a horse pulls a cart, what force accelerates the cart-horse system forward?

- (A) The force of the horse on the cart (B) The force the ground exerts forward on the horse's hooves
(C) The force the cart exerts on the horse (D) Gravity

Q.23 The dimensional formula of linear momentum is:

- (A) $[\text{MLT}]^{-1}$ (B) $[\text{MLT}]^{-2}$
(C) $[\text{ML}^2\text{T}^{-2}]$ (D) $[\text{ML}^{-1}\text{T}]$

Q.24 If the velocity of a moving object is doubled, what happens to its linear momentum?

- (A) Remains unchanged (B) Doubles
(C) Quadruples (D) Halves

Q.25 Why do passengers jerk backward when a stationary bus accelerates forward suddenly?

- (A) Inertia of rest (B) Inertia of motion
(C) Centripetal force (D) Friction with the roof

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET F)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & RAPID EXPLANATIONS (SET F)

Q. No.	Correct	Scientific Rationale & Formula Summary
Q.1	(C)	Newton defined mass as the quantitative measure of a body's inertia (resistance to change in motion).
Q.2	(B)	Momentum $p = mv$, having units $[\text{kg}] \times [\text{m/s}] = \text{kg} \cdot \text{m/s}$ (or $\text{N} \cdot \text{s}$).
Q.3	(B)	Newton's Second Law states that force equals rate of change of momentum: $F = \frac{\Delta p}{\Delta t} = ma$.
Q.4	(B)	$F = ma = 5 \text{ kg} \times 4 \text{ m/s}^2 = 20 \text{ N}$.
Q.5	(B)	Action and reaction are simultaneous, equal in magnitude, opposite in direction, and act on two distinct interacting bodies.
Q.6	(B)	Extending impact time Δt drastically reduces peak impact force ($F = \Delta p / \Delta t$), preventing hand injury.
Q.7	(C)	Due to inertia of direction, passengers tend to maintain their original straight-line direction of travel.
Q.8	(A)	Dust particles tend to remain stationary due to inertia of rest while the carpet fabric is suddenly jerked away.
Q.9	(A)	By conservation of momentum: $M V + m v = 0 \Rightarrow 4 V + (0.02 \times 400) = 0 \Rightarrow 4V = -8 \Rightarrow V = -2.0 \text{ m/s}$.
Q.10	(D)	Linear momentum is defined algebraically as $p = mv$.
Q.11	(B)	$1 \text{ N} = 1 \text{ kg} \times 1 \text{ m/s}^2 = 1 \text{ kg} \cdot \text{m/s}^2$.
Q.12	(B)	High-speed exhaust gas ejected downward produces an equal and opposite forward reaction thrust, conserving momentum.
Q.13	(B)	Newton's First Law: In the absence of an unbalanced net force, a body maintains constant velocity.
Q.14	(B)	The foot pushes backward on the ground; static friction pushes forward on the foot, propelling the person.
Q.15	(B)	Velocity is zero ($v = 0$); therefore, momentum $p = mv = 0$.
Q.16	(B)	Soft foam increases deceleration time Δt , drastically reducing peak impact force on the athlete's body.
Q.17	(A)	Impulse $J = F \Delta t = \Delta p$ (change in momentum).
Q.18	(C)	The Law of Conservation of Linear Momentum states that total system momentum remains invariant without external forces.
Q.19	(B)	Inertia is directly proportional to mass. Greater mass equals greater inertia.
Q.20	(B)	$\Delta p = F \times \Delta t = 10 \text{ N} \times 3 \text{ s} = 30 \text{ N} \cdot \text{s} = 30 \text{ kg} \cdot \text{m/s}$.
Q.21	(B)	Newton's First Law: Zero force is required to maintain uniform velocity in the absence of resistive friction.
Q.22	(B)	The external forward reaction force exerted by the ground on the horse's hooves accelerates the system forward.
Q.23	(A)	$\text{Mass} \times \text{Velocity} = [M] \times [L T^{-1}] = [MLT^{-1}]$.
Q.24	(B)	$p = mv$. If v doubles, momentum doubles.
Q.25	(A)	The lower body moves forward with the bus while the upper body tends to remain at rest due to inertia of rest.